1. **Check whether a number is positive or negative**

   Code:

```
package Day1;
import java.util.Scanner;

public class PosNeg {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter a number: ");
      int num = sc.nextInt();

      if(num >= 0)
         System.out.println("Output: Positive");
      else
         System.out.println("Output: Negative");

      sc.close();
   }
}
```

2. **Check whether a number is even or odd**

   Code:

```
package Day1;
import java.util.Scanner;

public class EvenOdd {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter a number: ");
      int num = sc.nextInt();

      if(num % 2 == 0)
         System.out.println("Output: Even");
      else
         System.out.println("Output: Odd");

      sc.close();
   }
}
```

3. **Find the largest of two numbers**

   Code:

```
package Day1;
import java.util.Scanner;

public class LargestTwo {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter first number: ");
      int a = sc.nextInt();

      System.out.print("Enter second number: ");
```

```
        int b = sc.nextInt();

        if(a > b)
            System.out.println("Output: " + a);
        else
            System.out.println("Output: " + b);

        sc.close();
    }
}
```

4. **Find the largest of three numbers**

Code:

```
package Day1;
import java.util.Scanner;

public class LargestThree {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        System.out.print("Enter third number: ");
        int c = sc.nextInt();

        if(a >= b && a >= c)
            System.out.println("Output: " + a);
        else if(b >= a && b >= c)
            System.out.println("Output: " + b);
        else
            System.out.println("Output: " + c);

        sc.close();
    }
}
```

5. **Check whether a year is a leap year**

Code:

```
package Day1;
import java.util.Scanner;

public class LeapYear {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter year: ");
        int year = sc.nextInt();

        if((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0))
```

```
            System.out.println("Output: Leap Year");
        else
            System.out.println("Output: Not Leap Year");

        sc.close();
    }
}
```

6. **Check whether a character is a vowel or consonant**

Code:

```
package Day1;
import java.util.Scanner;

public class VowelConsonant {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a character: ");
        char ch = Character.toLowerCase(sc.next().charAt(0));

        if(ch=='a'||ch=='e'||ch=='i'||ch=='o'||ch=='u')
            System.out.println("Output: Vowel");
        else
            System.out.println("Output: Consonant");

        sc.close();
    }
}
```

7. **Check eligibility for voting based on age**

Code:

```
package Day1;
import java.util.Scanner;

public class Voting {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter age: ");
        int age = sc.nextInt();

        if(age >= 18)
            System.out.println("Output: Eligible");
        else
            System.out.println("Output: Not Eligible");

        sc.close();
    }
}
```

8. **Check whether a number is divisible by 5 and 11**

Code:

```
package Day1;
import java.util.Scanner;
```

```
public class Divisible5And11 {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter number: ");
      int num = sc.nextInt();

      if(num % 5 == 0 && num % 11 == 0)
         System.out.println("Output: Divisible");
      else
         System.out.println("Output: Not Divisible");

      sc.close();
   }
}
```

9. **Find grade based on marks (if-else-if ladder)**

Code:

```
package Day1;
import java.util.Scanner;

public class Grade {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter marks: ");
      int marks = sc.nextInt();

      if(marks >= 90)
         System.out.println("Output: A");
      else if(marks >= 80)
         System.out.println("Output: B");
      else if(marks >= 70)
         System.out.println("Output: C");
      else if(marks >= 60)
         System.out.println("Output: D");
      else
         System.out.println("Output: F");

      sc.close();
   }
}
```

10. **Check whether a number is zero, positive, or negative**

Code:

```
package Day1;
import java.util.Scanner;

public class ZeroPosNeg {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter number: ");
      int num = sc.nextInt();
```

```
        if(num == 0)
            System.out.println("Output: Zero");
        else if(num > 0)
            System.out.println("Output: Positive");
        else
            System.out.println("Output: Negative");

        sc.close();
    }
}
```

11. **Find the smallest of three numbers**

Code:

```
package Day1;
import java.util.Scanner;

public class SmallestThree {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        System.out.print("Enter third number: ");
        int c = sc.nextInt();

        if(a <= b && a <= c)
            System.out.println("Output: " + a);
        else if(b <= a && b <= c)
            System.out.println("Output: " + b);
        else
            System.out.println("Output: " + c);

        sc.close();
    }
}
```

12. **Check whether a person is eligible for a loan (age + salary condition)**

Code:

```
package Day1;
import java.util.Scanner;

public class LoanEligibility {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter age: ");
        int age = sc.nextInt();

        System.out.print("Enter salary: ");
        double salary = sc.nextDouble();
```

```
        if(age >= 21 && salary >= 25000)
            System.out.println("Output: Eligible");
        else
            System.out.println("Output: Not Eligible");

        sc.close();
    }
}
```

13. Check whether a number is divisible by 3 or 7

Code:

```
package Day1;
import java.util.Scanner;

public class Divisible3Or7 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number: ");
        int num = sc.nextInt();

        if(num % 3 == 0 || num % 7 == 0)
            System.out.println("Output: Divisible");
        else
            System.out.println("Output: Not Divisible");

        sc.close();
    }
}
```

14. Simple calculator using switch case

Code:

```
package Day1;
import java.util.Scanner;

public class Calculator {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        System.out.print("Enter operator (+,-,*,/): ");
        char op = sc.next().charAt(0);

        switch(op) {
            case '+': System.out.println("Output: " + (a+b)); break;
            case '-': System.out.println("Output: " + (a-b)); break;
            case '*': System.out.println("Output: " + (a*b)); break;
```

```
            case '/': System.out.println("Output: " + (a/b)); break;
            default: System.out.println("Invalid Operator");
        }

        sc.close();
    }
}
```

15. **Display day of week using switch case**

Code:

```
package Day1;
import java.util.Scanner;

public class WeekDay {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter day number (1-7): ");
        int day = sc.nextInt();

        switch(day) {
            case 1: System.out.println("Output: Monday"); break;
            case 2: System.out.println("Output: Tuesday"); break;
            case 3: System.out.println("Output: Wednesday"); break;
            case 4: System.out.println("Output: Thursday"); break;
            case 5: System.out.println("Output: Friday"); break;
            case 6: System.out.println("Output: Saturday"); break;
            case 7: System.out.println("Output: Sunday"); break;
            default: System.out.println("Output: Invalid");
        }

        sc.close();
    }
}
```